

HEALTHCONNECT CLINIC

Final Analytics & Decision Support Report

Data Analytics Track

Executive Summary

HealthConnect is focused on reducing missed appointments, improving patient appointment attendance, making better use of available appointment slots, and providing more effective administrative support. The Data Analytics track supports these objectives by analysing appointment data and developing a Power BI decision-support dashboard that highlights observed attendance and no-show patterns.

Across the project, the analytics work progressed from initial analysis and dashboard development through advanced analysis, testing, refinement, and final integration.

The final analytics component provides visibility into appointment attendance and no-show patterns across booking lead time, previous no-show history, reminder channels, appointment characteristics, and other relevant segments. These findings can support targeted administrative attention, appointment planning, and better use of available appointment capacity.

Week 7 → Week 8 Transition

Item	Final Documentation
Main Week 7 output	HealthConnect Analytics Testing & Refinement Report, supported by the Week 7 testing/validation record and refined Power BI dashboard.
Most important testing result	Important dashboard outputs, calculations, filters, and analytical components were tested and documented during Week 7.
Major issue identified	Week 7 focused on identifying and addressing issues found during KPI, visual, filter, and analytical validation.
Major refinement	The dashboard was refined to improve clarity, usability, and business interpretation while retaining the validated analytical story.
Ready component	Final HealthConnect Data Analytics and Decision Support dashboard and supporting findings.
Presentation demonstration	Problem → data → analytics → findings → business value → limitations → wider solution.

Final Integration Readiness

Readiness Question	Response
Final component	Refined Power BI dashboard, final KPIs, validated findings, business insights, recommendations, limitations, executive summary, and presentation materials.
Problem addressed	Understanding appointment attendance and no-show patterns to support HealthConnect' business objectives.
Current status	Prepared for final integration, presentation, and project showcase.
Week 7 improvement	Testing and refinement were used to validate important outputs, calculations, filters, findings, and dashboard usability.
Tracks that may use analytics	Data Science may use validated findings for model interpretation; Project Management can use final insights for business recommendations and decisions.
Output provided	Final KPIs, validated findings, visualizations, insights, recommendations, limitations, executive summary, and presentation materials.
Readiness evidence	Final dashboard, KPI checks, testing evidence, refined visuals, findings, screenshots, report, and presentation.
Remaining limitations	Observed relationships do not establish causation; segment volume matters; formal predictive validation is not claimed where it was not completed.

Final Analytics Component

The final analytics component brings the HealthConnect work together into a decision-support dashboard and supporting documentation. The dashboard should move from overall performance to the segments and interactions most useful for decision-making.

Final KPI Summary

KPI	Purpose
No-Show Rate	Proportion of appointments recorded as no-shows.
Attendance Rate	Proportion of appointments resulting in attendance.
Cancellation Rate	Proportion of appointments recorded as cancellations.
Reminder-Associated Attendance Rate	Attendance associated with reminder

	categories.
Repeat No-Show Rate	No-show behaviour among appointments/patients with previous no-show history according to the project definition.

Validated Key Findings

Booking lead time: No-show rates varied across booking lead-time groups, with higher observed rates in longer booking-lead categories.

Previous no-show history: Groups with previous no-show history generally showed higher subsequent observed no-show rates.

Combined segmentation: Booking Lead Group $\times$ Previous No-Show Group provides a more detailed view of observed no-show patterns.

Reminder channel: Reminder channel categories showed different observed attendance/no-show patterns; these are associations, not proof of causation.

Waiting time: Longer waiting time categories showed differences in observed no-show rates; segment volume should be considered.

Key Visualizations

Visualization	Purpose
KPI Cards	Overall appointment performance.
No-Show Rate by Booking Lead Group	Variation in observed no-show rates across booking lead segments.
No-Show Rate by Previous No-Show Group	Observed pattern across previous no-show history.
Booking Lead Group $\times$ Previous No-Show Group Matrix	Combined segment analysis.
Previous No-Show Group $\times$ Reminder Channel Matrix	Observed pattern across history and reminder channel.
Supporting visuals	Appointment type, reminder channel, waiting time, and other relevant context.

Business Insights

- Longer booking-lead categories should be monitored because higher observed no-show rates were seen in these segments.
- Previous no-show history is an important segmentation variable because groups with prior no-shows generally showed higher subsequent observed no-show rates.
- Combining booking lead time with previous no-show history gives a more detailed basis for identifying segments for administrative attention.

- Reminder-channel differences provide an opportunity to monitor reminder performance by channel over time.
- Waiting-time patterns should be reviewed alongside appointment volume so that small-sample percentages are not overinterpreted.

Actionable Recommendations

Recommendation	Action
Review longer booking-lead appointments	Consider reviewing reminder timing and administrative follow up for appointments booked further in advance.
Use previous no-show history for targeted attention	Consider appropriate administrative follow up for patients with previous no-show history without assuming individual behaviour.
Monitor reminder channel performance	Track attendance/no-show outcomes by reminder channel over time.
Use combined segmentation	Review booking lead time and previous no-show history together when prioritizing appointment management attention.
Monitor waiting-time patterns with volume	Include appointment counts when reviewing waiting time segments.

Business Value and Decision Support

The analytics component provides HealthConnect with evidence-based visibility into appointment attendance and no-show patterns. By identifying observed patterns across booking behaviour, previous no-show history, reminder channels, and other appointment characteristics, the analysis can support targeted administrative attention, appointment planning, and better utilisation of available appointment slots.

The dashboard also provides a common visual reference for discussing appointment performance. Decision makers can examine segments and interactions rather than relying only on overall totals.

Proposed Data Science handoff: Booking Lead Time, Previous No-Show History, Reminder Channel, and the interaction between Booking Lead Time and Previous No-Show History. These are candidate inputs for future predictive testing, not completed model validation.

Analytical Limitations

- Observed relationships do not establish causation.
- Rates should be interpreted together with appointment volume, particularly for small segments.
- Reminder channel differences do not prove that a channel caused higher attendance.
- Segment-level patterns may change with another dataset or time.
- The analytics component does not provide predictive model performance or clinical conclusions.
- Operational actions should be monitored after implementation.

Conclusion

Week 8 Data Analytics deliverable consolidates the HealthConnect analytics work into a final decision-support package. It combines validated KPIs, analytical findings, dashboard visualizations, business insights, recommendations, limitations, and presentation materials. The final package communicates what the data shows and how the findings can support HealthConnect's operational objectives.